Practical 7: ARIMA Modeling of Commercial Bank Real Estate Loans

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1. Question

Consider the monthly volume of commercial bank real estate loans (in billions of dollars) data set stored in a text file. Write the R program for the following:

(f) Import data into the R environment.

(g) Convert the data into a time series object.

(h) Plot the data to identify the dominant component.

(i) Check stationarity or non-stationarity using ACF/PACF plot.

(j) Check stationarity or non-stationarity using the Augmented Dickey-Fuller (ADF) test.

(k) If data is non-stationary, make it stationary using an appropriate operator.

(l) Based on the dominating component, select the suitable model to fit the data and finalize the order of the model using ACF/PACF plots with model selection criteria such as AIC.

(m) Fit the data using the selected model and estimate the parameters of the model.

(n) Check the goodness of fit of the model.

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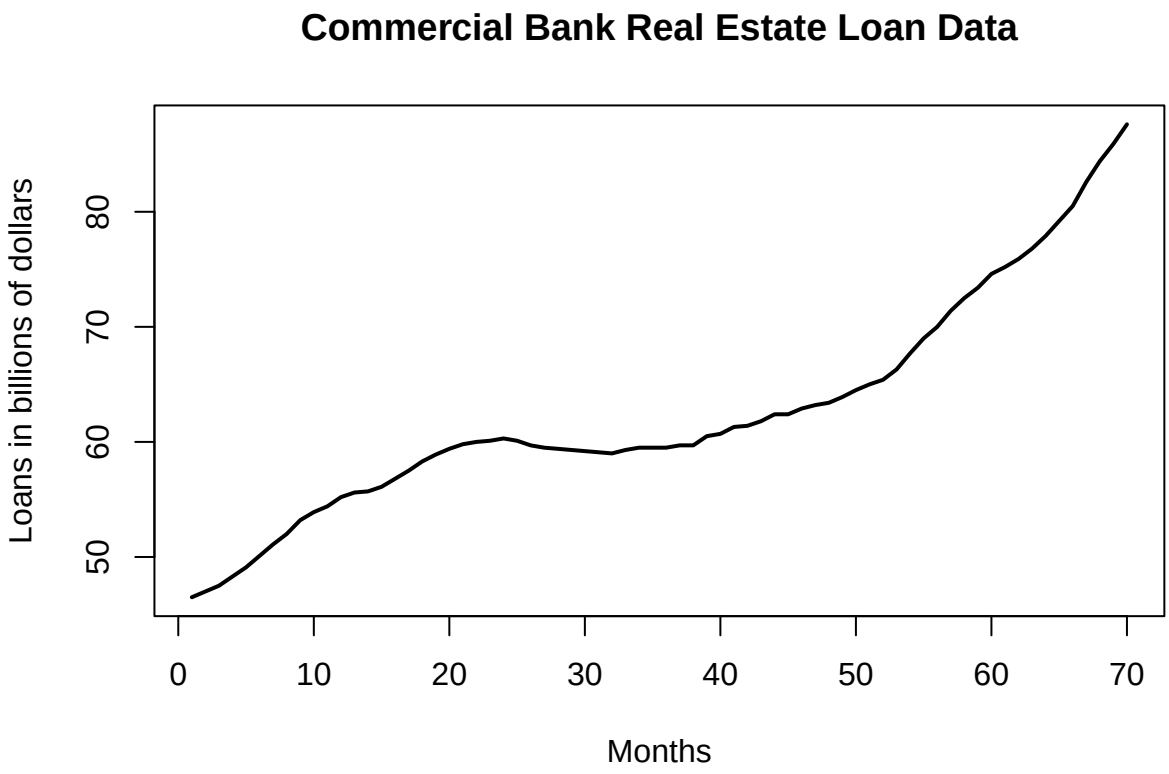
2. R Code (Question-wise Solution)

```r
# To delete previous history or stored variables
rm(list = ls(all = TRUE))

library(tseries)
library(forecast)
```

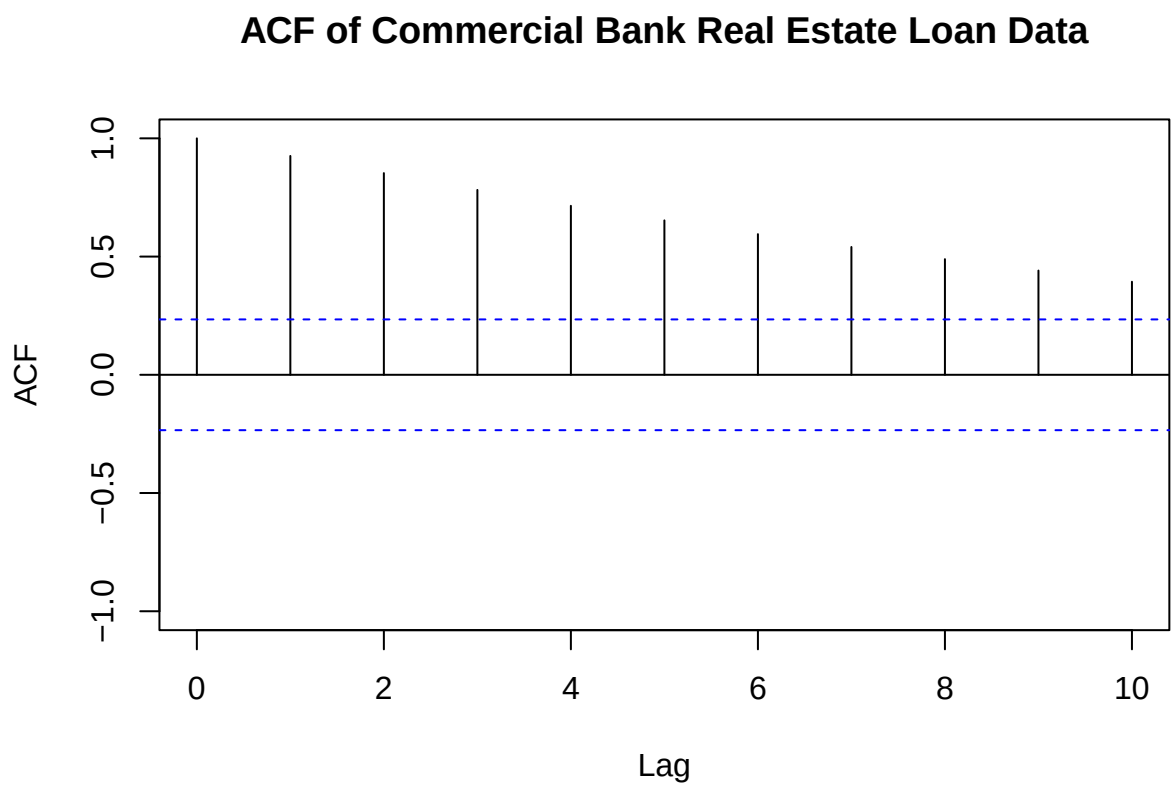
```r
# ------------------------------------------------
# (f) Import data into the R environment
# ------------------------------------------------

bank_data <- as.ts(scan("bank_case.txt"))

# Printing the data
bank_data
```

```
## Time Series:
## Start = 1
## End = 70
## Frequency = 1
##  [1] 46.5 47.0 47.5 48.3 49.1 50.1 51.1 52.0 53.2 53.9 54.4 55.2 55.6 55.7 56.1
## [16] 56.8 57.5 58.3 58.9 59.4 59.8 60.0 60.1 60.3 60.1 59.7 59.5 59.4 59.3 59.2
## [31] 59.1 59.0 59.3 59.5 59.5 59.5 59.7 59.7 60.5 60.7 61.3 61.4 61.8 62.4 62.4
## [46] 62.9 63.2 63.4 63.9 64.5 65.0 65.4 66.3 67.7 69.0 70.0 71.4 72.5 73.4 74.6
## [61] 75.2 75.9 76.8 77.9 79.2 80.5 82.6 84.4 85.9 87.6
```

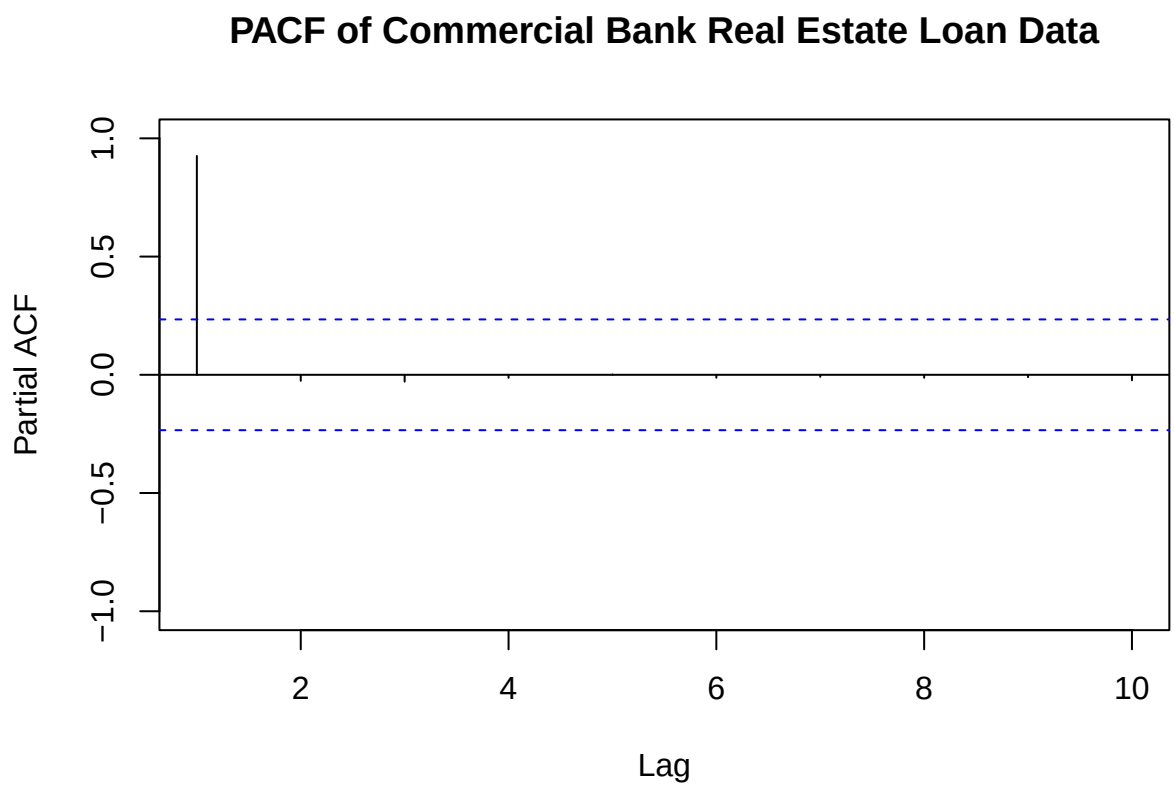
```r
# ------------------------------------------------
# (g) Convert data into time series object
# ------------------------------------------------
# (Already converted using as.ts())



# ------------------------------------------------
# (h) Plot the data to identify dominant component
# ------------------------------------------------

plot(bank_data,
     main = "Commercial Bank Real Estate Loan Data",
     xlab = "Months",
     ylab = "Loans in billions of dollars",
     lwd = 2)
```




```r
# ------------------------------------------------
# (i) Check stationarity using ACF/PACF plot
# ------------------------------------------------

acf(bank_data,
    main = "ACF of Commercial Bank Real Estate Loan Data",
    lag.max = 10,
    ylim = c(-1,1))
```




```r
pacf(bank_data,
     main = "PACF of Commercial Bank Real Estate Loan Data",
     lag.max = 10,
     ylim = c(-1,1))
```




```r
# ---------------------------------------------------
# (j) Check stationarity using ADF Test
# ---------------------------------------------------

adf.test(bank_data)
```

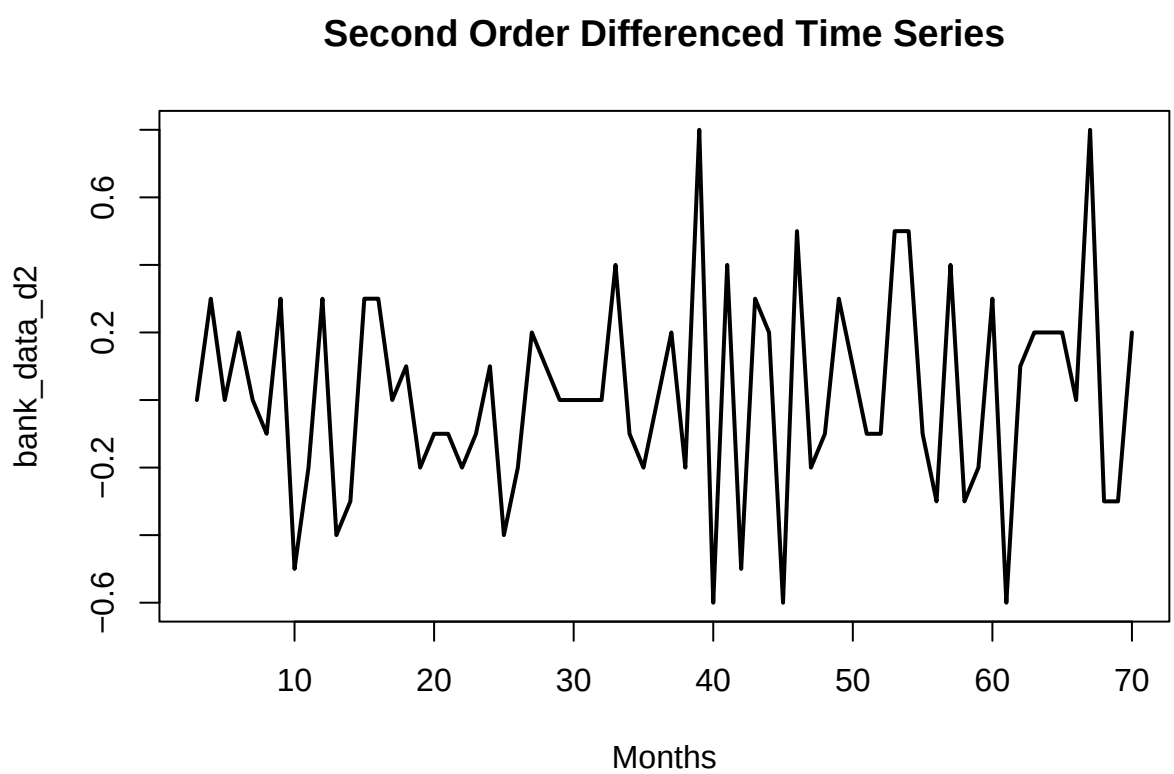
```
##
##  Augmented Dickey-Fuller Test
##
## data:  bank_data
## Dickey-Fuller = -0.26816, Lag order = 4, p-value = 0.9894
## alternative hypothesis: stationary
```

```r
# ---------------------------------------------------
# (k) If non-stationary, make it stationary
# Using difference operator
# ---------------------------------------------------

# First differencing
bank_data_d1 <- diff(bank_data)

adf.test(bank_data_d1)
```

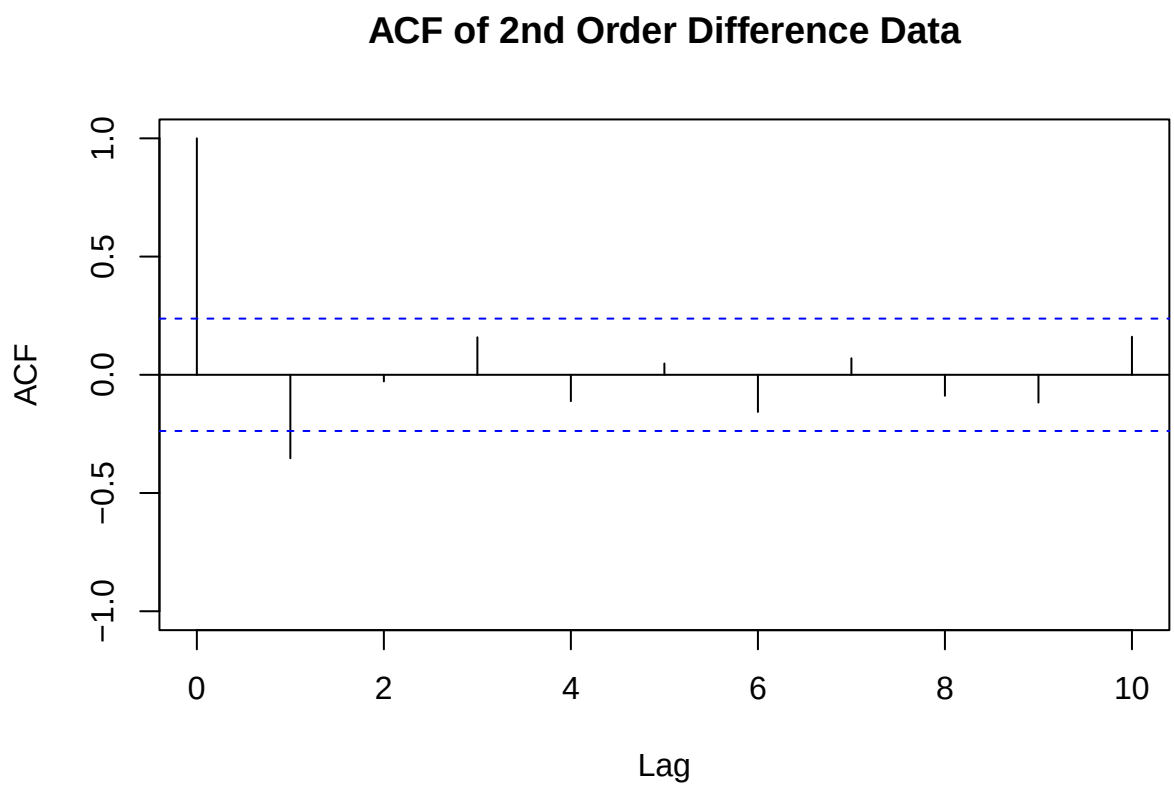
```
##
##  Augmented Dickey-Fuller Test
```

```
##
## data:  bank_data_d1
## Dickey-Fuller = -1.7533, Lag order = 4, p-value = 0.6755
## alternative hypothesis: stationary
```

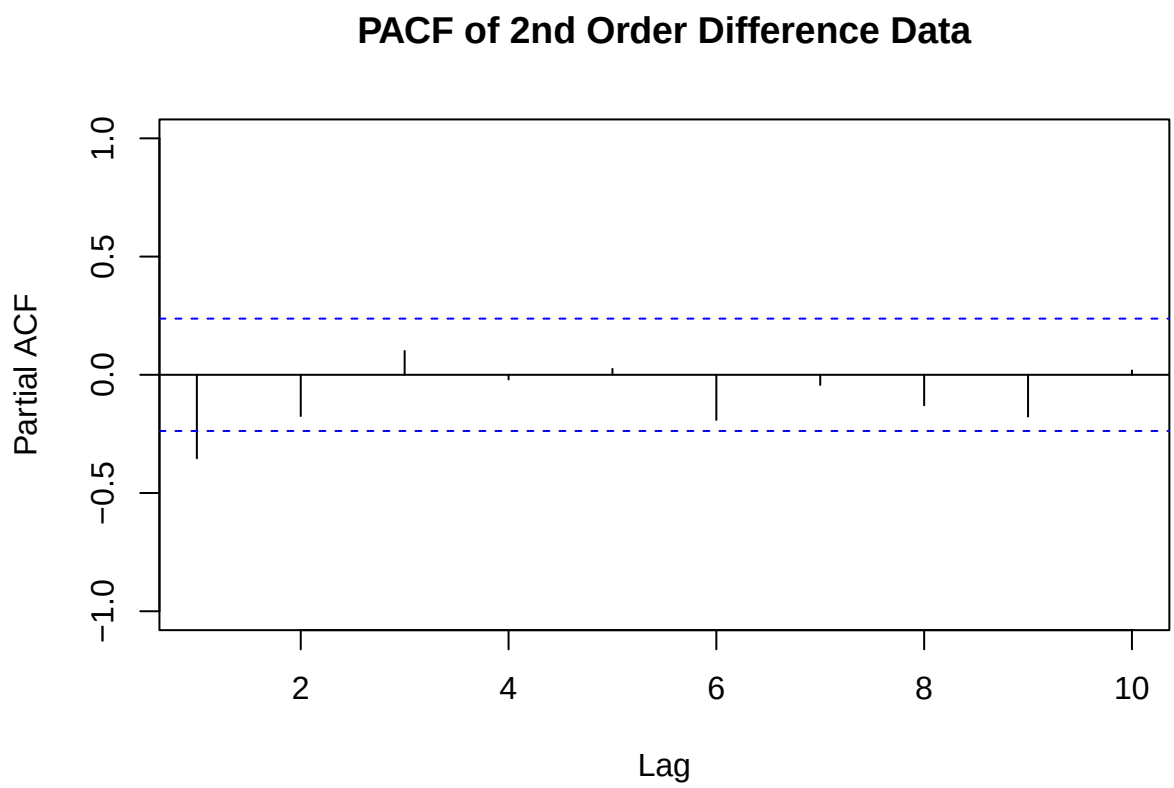
```r
# Second differencing (since first was not stationary)
bank_data_d2 <- diff(bank_data_d1)

plot(bank_data_d2,
     main = "Second Order Differenced Time Series",
     xlab = "Months",
     lwd = 2)
```




```r
acf(bank_data_d2,
    main = "ACF of 2nd Order Difference Data",
    lag.max = 10,
    ylim = c(-1,1))
```




```r
pacf(bank_data_d2,
     main = "PACF of 2nd Order Difference Data",
     lag.max = 10,
     ylim = c(-1,1))
```




```r
adf.test(bank_data_d2)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  bank_data_d2
## Dickey-Fuller = -4.0531, Lag order = 4, p-value = 0.01265
## alternative hypothesis: stationary
```

```r
# ---------------------------------------------------
# (l) Model Identification using ACF/PACF and AIC
# From ACF/PACF of second difference:
# Possible p = 0,1,2
# Possible q = 0,1
# Best model selected: ARIMA(0,2,1)
# ---------------------------------------------------

bank_data_fit <- arima(bank_data, order = c(0,2,1))

bank_data_fit
```

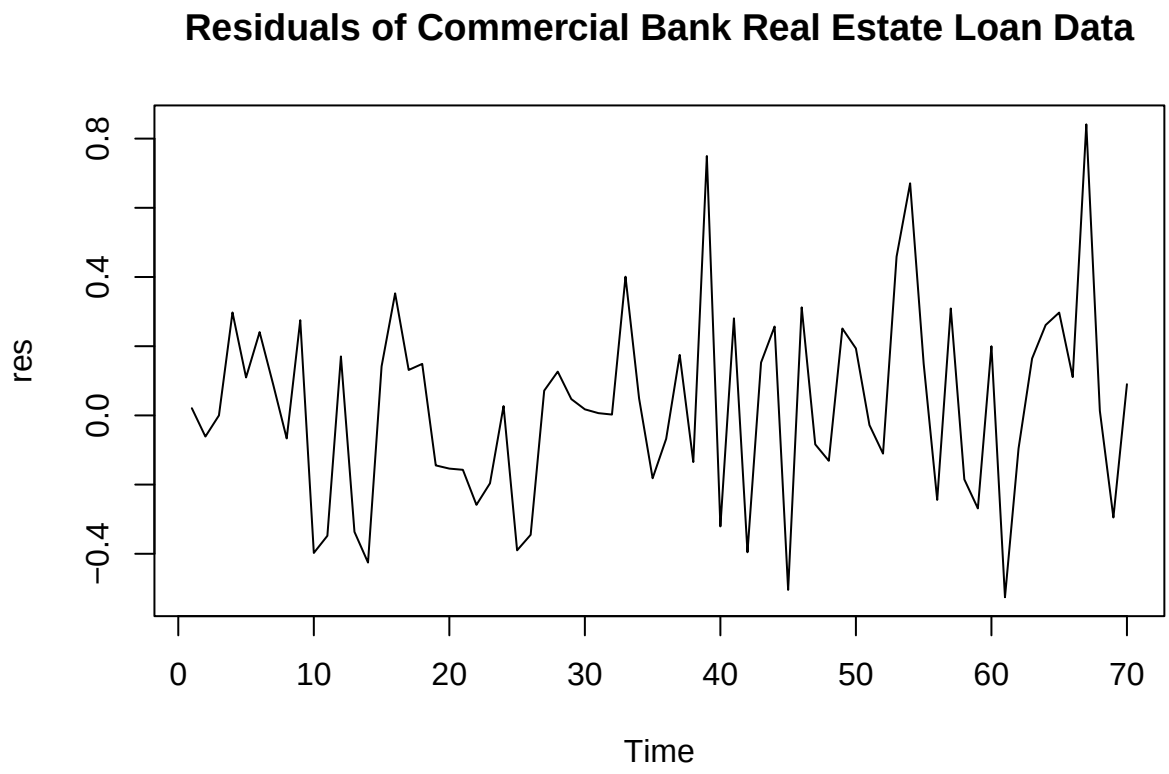
```
##
## Call:
## arima(x = bank_data, order = c(0, 2, 1))
##
```

```
## Coefficients:
##           ma1
##       -0.3723
## s.e.   0.1074
##
## sigma^2 estimated as 0.08089:  log likelihood = -11.07,  aic = 26.13
```

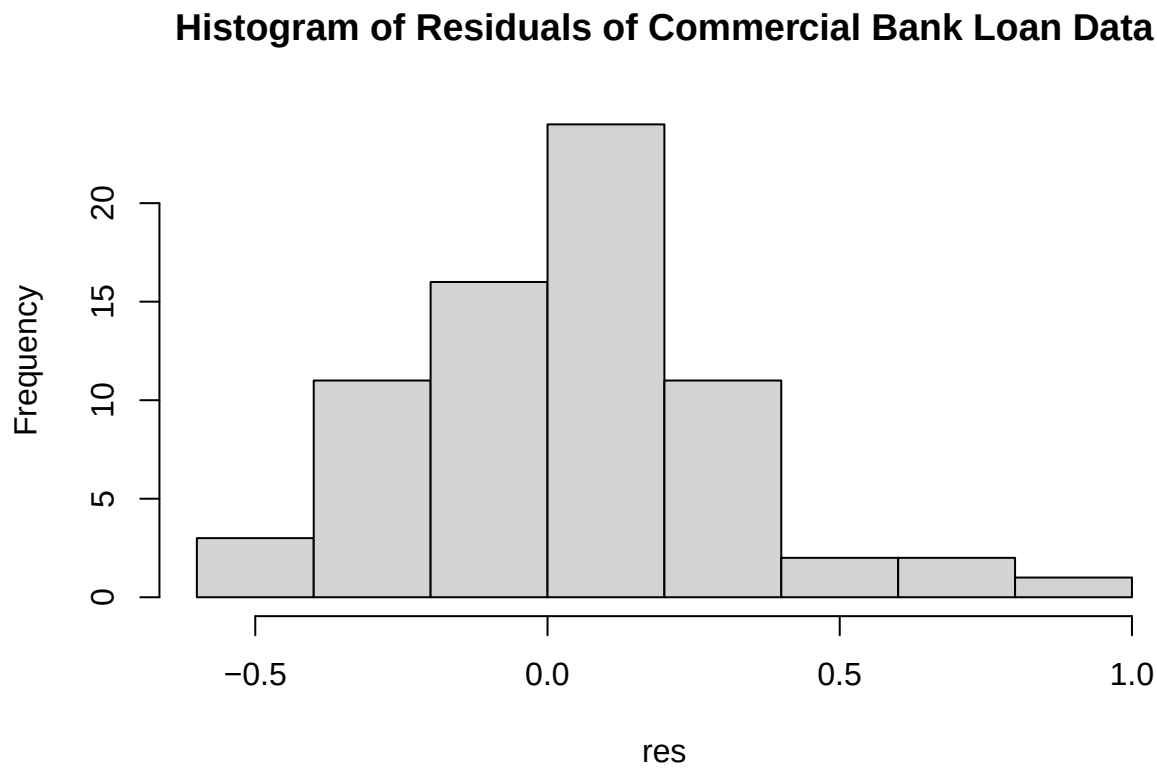
```r
# ------------------------------------------------
# (m) Fit the data and estimate parameters
# (Already fitted above using arima function)
# ------------------------------------------------



# ------------------------------------------------
# (n) Check goodness of fit (Diagnostic Checking)
# ------------------------------------------------

# Extract residuals
res <- bank_data_fit$residuals
res
```

```
## Time Series:
## Start = 1
## End = 70
## Frequency = 1
##  [1]  0.0207954298 -0.0612681076 -0.0000477981  0.2974844128  0.1097007931
##  [6]  0.2407549685  0.0896135857 -0.0666386102  0.2751913095 -0.3975500319
## [11] -0.3480024676  0.1704434072 -0.3365462337 -0.4252915933  0.1416697143
## [16]  0.3527417111  0.1313209497  0.1488890065 -0.1445706444 -0.1538216880
## [21] -0.1572657259 -0.2585478945 -0.1962538707  0.0269373486 -0.3899715909
## [26] -0.3451811284  0.0714937526  0.1266161534  0.0471374748  0.0175486418
## [31]  0.0065331211  0.0024321923  0.4009054722  0.0492516640 -0.1816642742
## [36] -0.0676311427  0.1748218438 -0.1349162012  0.7497725301 -0.3208698671
## [41]  0.2805445051 -0.3955570899  0.1527394662  0.2568628294 -0.5043734562
## [46]  0.3122286104 -0.0837615276 -0.1311832795  0.2511622462  0.1935042941
## [51] -0.0279609786 -0.1104094927  0.4588960450  0.6708407668  0.1497449091
## [56] -0.2442520035  0.3090682954 -0.1849380700 -0.2688499325  0.1999108205
## [61] -0.5255759158 -0.0956647774  0.1643853023  0.2611984161  0.2972406239
## [66]  0.1106586485  0.8411967123  0.0131661146 -0.2950984343  0.0901388596
```

```r
# Plot residuals
plot(res,
     main = "Residuals of Commercial Bank Real Estate Loan Data")
```




```r
# Check Normality of Errors
hist(res,
     main = "Histogram of Residuals of Commercial Bank Loan Data")
```



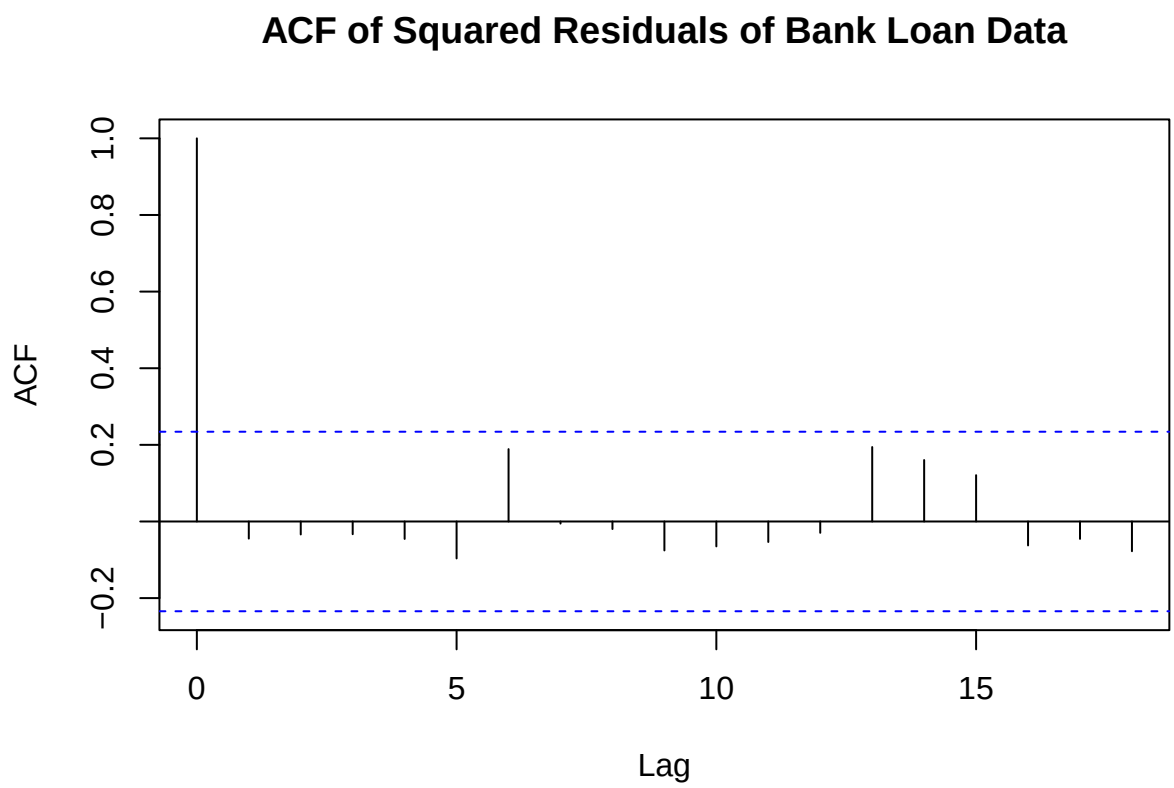

```r
shapiro.test(res)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res
## W = 0.97676, p-value = 0.218
```

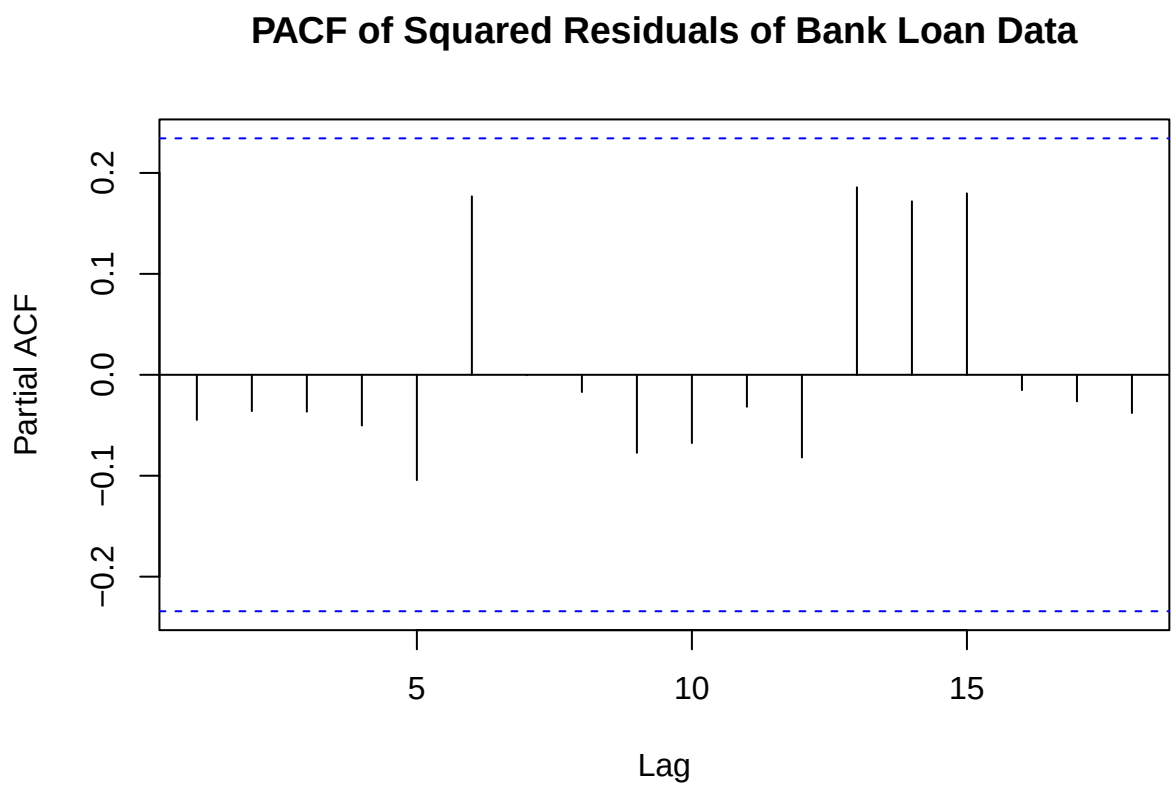
```r
# Check Serial Correlation
Box.test(res,
         lag = 10,
         type = "Box-Pierce")
```

```
##
##  Box-Pierce test
##
## data:  res
## X-squared = 8.4829, df = 10, p-value = 0.5818
```

```r
# Check Heteroscedasticity
acf(res^2,
    main = "ACF of Squared Residuals of Bank Loan Data")
```




```
pacf(res^2,
     main = "PACF of Squared Residuals of Bank Loan Data")
```



Conclusion

The original data showed a clear upward trend and was non-stationary based on ACF and ADF test.

After first differencing, the series remained non-stationary.
After second differencing, the series became stationary.

Based on ACF/PACF plots of second differenced data, ARIMA(0,2,1) was selected.

Diagnostic checking confirmed:

- Errors are normally distributed
- No serial correlation
- Constant variance

Hence, ARIMA(0,2,1) is a good fit for the data.